

Computer Crime and Security

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Computers Are Tools

Computers assist us in our work, expand our thinking, and provide entertainment.

Computers Are also Used to Commit Crimes

Preventing, detecting, and prosecuting (act against) computer crime is a challenge.

Cybercrime and the Law

Computer-related crime

Any illegal act for which knowledge of computer technology is involved for its investigation, perpetration, or prosecution.



Types of Computer Crimes

- The crimes that are typically mentioned in lists of “computer crimes” are:
- hacking, viruses, pirating, illegal trading fraud, scams, money laundering, libel (defamation), cyber stalking (Cyberstalking is a crime in which someone harasses or stalks a victim using electronic or digital means, such as social media, email, instant messaging (IM), cyber terrorism (the politically motivated use of computers and information technology to cause severe disruption or widespread fear in society), identity theft and fraud,

Types of Computer Crimes

and wireless theft. ("Wireless theft" refers to the act of stealing personal information or data through a wireless connection, like Wi-Fi, typically by exploiting vulnerabilities in an unsecured network to access sensitive details without authorization, often including credit card numbers, login credentials, or other sensitive data).



Internet criminals

- Because of the global scope of the Internet, criminals sitting alone anywhere in the world can move their fingers over keyboards,
- stare at computer screens, and steal from those in distant places,
- and they can cause destruction on the lives of people across the globe by launching viruses or denial-of-service attacks (shut down machine for users).

The Advance of Technology

- Technology and criminology have always been closely linked, Technology and criminology is the study of crime and criminal behavior in the context of technology.
- Technology can be used by crime fighters and lawbreakers
- As technology progresses, it facilitates new forms of behavior, including new crimes

High Technology and Criminal Opportunity

- Routes to illegitimate access to computerized information
 - Direct access
 - Computer trespass (unlawful access)
- **Cybercrime**
 - any violation of a federal or state computer-crime statute.

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What is Cyber Crime?

The adopted definition of Cyber Crime is:

- **Cyber Dependent Crimes**, where a digital system is the target as well as the means of attack. These include attacks on computer systems to disrupt IT infrastructure, and stealing data over a network using malware (the purpose of the data theft is usually to commit further crime).
- **Cyber Enabled Crimes**. 'Existing' crimes that have been transformed in scale or form by their use of the Internet. The growth of the Internet has allowed these crimes to be carried out on an industrial scale.
- The use of the Internet to facilitate drug dealing, people smuggling and many other 'traditional' crime types.

What is Cyber Crime?

Cyber Dependent Crimes



ActionFraud
National Fraud & Cyber Crime Reporting Centre
0300 123 2040

Cyber Enabled Crimes



96%



High Technology and Criminal Opportunity

Types of cybercrime

- Internal cybercrimes
- Internet/telecommunications crimes
- Support of criminal enterprises
- Computer-manipulation crimes
- Hardware, software, and information theft

The Extent of Cybercrime

Software piracy

The unauthorized and illegal copying of software programs

Phishing

An Internet-based scam that uses official-looking e-mail messages to steal valuable information

May threaten the viability of e-commerce



The Extent of Cybercrime

Not all cybercrime is committed for financial gain

Criminal mischief

- creating/transmitting malicious forms of programming code
- Criminal mischief, more commonly known as vandalism (damage), is **the intentional and malicious destruction of another person's property**

Cybercrime and the Law

- Communications Decency Act (1996)
- No Electronic Theft Act (1997)
- Digital Theft Deterrence and Copyright Damages Improvement Act (1999)
- Cyber Security Enhancement Act (2002)

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Computer crime and abuse

- **Computer crime:** Commission of illegal acts through use of compute or against a computer system – computer may be object or instrument of crime
- **Computer abuse:** Commission of acts involving a computer that may not be illegal but are considered unethical, i.e. spamming



Pakistan and Cyber Crimes: Problems and Preventions

- Government is constantly trying to formulate the regulation for controlling the cyber crimes.

It is essential to make the internet more secure and protect the user of internet from cyber attacks

- Anti cybercrime law
- The organization are moving towards paper less economy where everything will be digital. They want to safe guard the sensitive information rather than the physical security of the system only.
- Electronic Transaction Ordinance (ETO).

The History and Nature of Hacking

Computer hacking may have started in the late 1950s with the creation of the interstate phone system and direct distance dialing

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The History and Nature of Hacking

- **Phone phreaks**
 - Used special telecommunications access codes and other restricted technical information to avoid paying long-distance charges
 - Modern version involves electronic theft of cell phone numbers and access codes
- **Voice mail hacking and voice mail fraud**

Malware

Malware, short for malicious software, refers to any intrusive software developed by cybercriminals (often called hackers) to steal data and damage or destroy computers and computer systems. Examples of common malware include viruses, worms

Types of malware

Different types of malware have the following unique traits and characteristics:

- Virus.** A virus is the most common type of malware that can execute itself and spread by infecting other programs or files.
- Worm.** A worm can self-replicate without a host program and typically spreads without any interaction from the malware authors.
- Trojan horse.** A [Trojan horse](#) is designed to appear as a legitimate software program to gain access to a system. Once activated following installation, Trojans can execute their malicious functions

How to detect malware

Users might be able to detect malware if they observe unusual activity. Common malware symptoms include the following:

- A sudden loss of disk space.
- Unusually slow computer or device speeds.
- A blue screen of death.
- Repeated system crashes or freezes.
- Changed browser settings and redirects.
- Increase in unwanted internet activity.